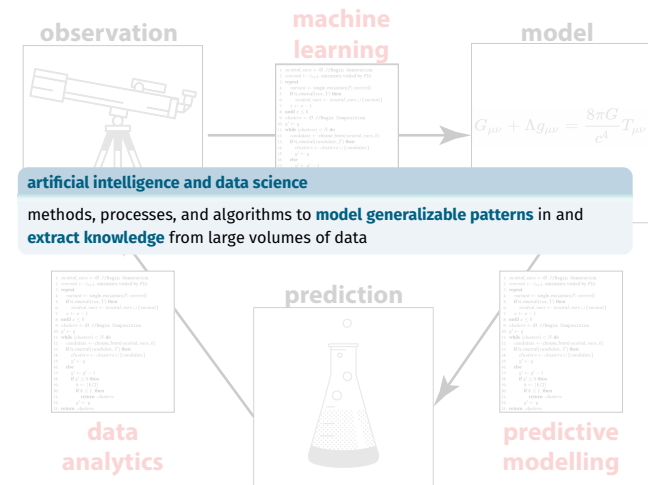


Algorithms, AI & Data Science II

Prof. Dr. Ingo Scholtes

Chair of Machine Learning for Complex Networks
Center for Artificial Intelligence and Data Science (CAIDAS)
Julius-Maximilians-Universität Würzburg
ingo.scholtes@uni-wuerzburg.de

Motivation



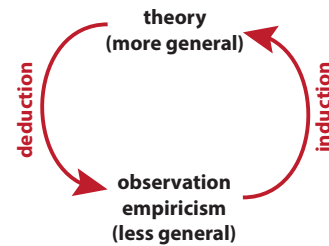
Notes:

- **Lecture L21: Summary and Outlook** 21.07.2025
- **Educational objective:** We summarize the course, give an outlook to advanced topics that you can study in follow-up courses and textbooks, and discuss the results of the lecture evaluation.
- **Handout version:** 2025/06/24 15:27:41

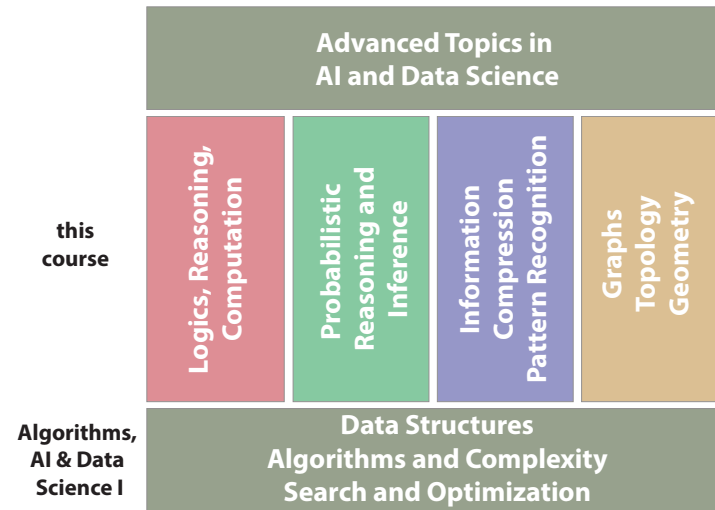
Notes:

Inductive vs. deductive reasoning

- ▶ extending the simplistic view given in the motivation, we considered both **inductive and deductive methods of knowledge extraction**
- ▶ **deductive approaches** include symbolic AI like, e.g., logical inference
- ▶ **inductive approaches** include statistical inference and machine learning
- ▶ inductive approaches to AI have recently become much more common (and successful)
- ▶ but: **both approaches have their justification in AI**



Algorithms, AI, and Data Science II



Notes:

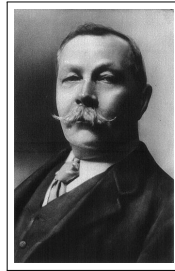
Notes:

Final syllabus

1/2

Part I - Logics, Reasoning, and Computation

- ▶ L01 - Motivation → 28.04.2025
- ▶ L02 - Formal Foundations of Computing → 30.04.2025
- ▶ L03 - Word Problems and Automata → 05.05.2025
- ▶ L04 - Turing Machines and Computability → 06.05.2025
- ▶ L05 - Propositional Logic and Boolean Algebra → 14.05.2025
- ▶ L06 - Satisfiability in Propositional Logic → 19.05.2025
- ▶ L07 - Logical Inference → 21.05.2025
- ▶ L08 - First-Order Logic → 26.05.2025



Sir Arthur Conan Doyle
1859 – 1930

Part II - From Logical to Probabilistic Reasoning

- ▶ L09 - Probabilistic Foundations → 28.05.2025
- ▶ L10 - Statistical Modelling and Inference → 11.06.2025
- ▶ L11 - Bayesian Inference → 16.06.2025
- ▶ L12 - Hypothesis Testing → 18.06.2025
- ▶ L13 - Challenges in Data Science → 23.06.2025

logical inference

“He was, I take it, the most perfect reasoning and observing machine that the world has seen [...]”
→ Sherlock Holmes, A Scandal in Bohemia

image credit: Wikimedia Commons, public domain

Notes:

Final syllabus

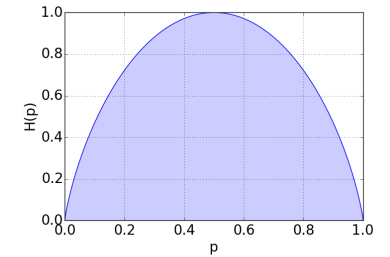
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Part III - Information, Compression, Pattern Recognition

- ▶ L14 - Markov Chains and Random Walks → 25.06.2025
- ▶ L15 - Markov Chain Convergence and PageRank → 30.06.2025
- ▶ L16 - Markov Chain Monte Carlo → 02.07.2025
- ▶ L17 - Information Theory and Data Compression → 07.07.2025
- ▶ L18 - Description Length Minimization → 09.07.2025

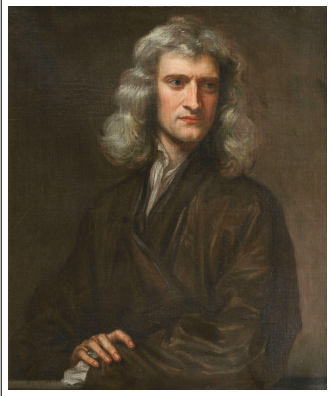
Part IV - Graphs and Topology

- ▶ L19 - Graphs, Laplacians and Minimum Cuts → 14.07.2025
- ▶ L20 - Flow Maximization in Graphs → 16.07.2025
- ▶ L21 - Summary and Conclusion → 21.07.2025



Notes:

Is that all there is to know?



Isaac Newton

“What we know is a drop. What we do not know is an ocean.”

Isaac Newton

image credit: Portrait of Isaac Newton by Godfrey Kneller, public domain

Notes:

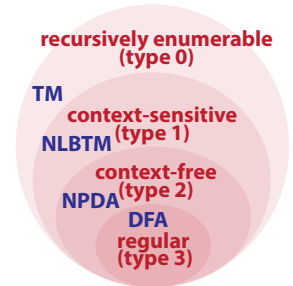
Outlook: Formal languages and NLP

- ▶ in L02 and L03 we introduced **formal languages and formal grammars**
- ▶ **Chomsky hierarchy** is foundation for **computational linguistics and natural language processing**
- ▶ we studied which automata models are needed to solve **word decision problems** for different formal languages

further topics

- ▶ regular expressions and **Kleene's theorem**
- ▶ **pumping lemma** for regular and context-free languages
- ▶ **Backus-Naur** form for context-free grammars
- ▶ **CYK algorithm** for context-free languages in Backus-Naur form

- ▶ see course **Theoretische Informatik** (Prof. Glasser)



Chomsky hierarchy with automata models that solve word decision problem → L03

Notes:

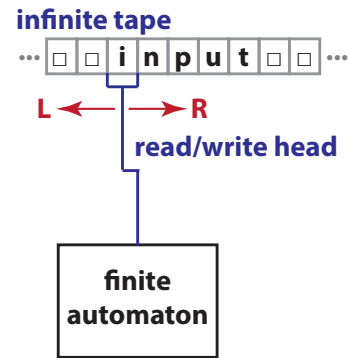
Outlook: Automata, computability, and complexity

- ▶ in L03 we explored different **automata models** to solve word decision problems
- ▶ in L04 we introduced (universal) **Turing machines** as basis for our notion of **computability**

further topics

- ▶ **LOOP, WHILE, GOTO-computability**
- ▶ **primitive recursive** and μ -recursive functions
- ▶ **Ackermann function**
- ▶ **Post's correspondence problem**
- ▶ **Gödel's incompleteness theorem**
- ▶ P, NP, **NP-completeness**, and reducibility

- ▶ see courses **Theoretische Informatik** and **Berechenbarkeitstheorie** (Prof. Glasser)



Turing machine → L03

Outlook: From first- to higher-order logic

- ▶ In L05 we introduced **propositional and first-order logic**
- ▶ in L06 and L08 we studied the **satisfiability problem** in propositional and first-order logic
- ▶ in L07 and L08 we explored **logical inference** in propositional and first-order logic

further topics

- ▶ **unification algorithm** for first-order inference
- ▶ **backward chaining** in first-order logic
- ▶ **higher-order and modal logic**
- ▶ **knowledge representation** and **ontologies**
- ▶ **logic programming** and **PROLOG**

- ▶ see course **Logik für Informatiker** (Prof. Seipel)

$$(X_0 \vee X_1 \vee X_2) \wedge (\neg X_1 \vee \neg X_2) \wedge X_0 \wedge \neg X_0$$

resolution algorithm

```
procedure RESOLUTION(F)
  choose X that also appears as ¬X in F
  for clauses ci with X ∈ ci do
    for clauses cj with ¬X ∈ cj do
      r ← ci - {X} ∨ cj - {¬X}
      F ← add_clause(F, r)
    end for
  end for
  for clauses ci with X ∈ ci or ¬X ∈ ci do
    F ← remove_clause(F, ci)
  end for
end procedure
```

→ L06

Notes:

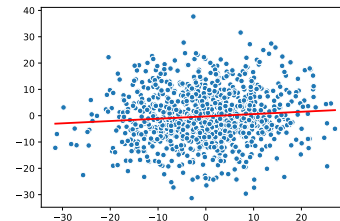
Notes:

Outlook: Pattern recognition and dimensionality reduction

- ▶ in L09 – L11 we introduced foundations of **statistical modelling** as well as **frequentist and Bayesian inference**
- ▶ in L11 we took a statistical inference perspective on **linear regression**
- ▶ in L12 – L13 we explored **hypothesis testing in data**

further topics

- ▶ **MCMC sampling** in Bayesian inference
- ▶ **expectation maximization** for inference with hidden variables
- ▶ cluster detection with **gaussian mixture model** and **DBSCAN**
- ▶ classification with **logistic regression** and **neural networks**
- ▶ dimensionality reduction with **principal component analysis** (PCA), **singular value decomposition** (SVD), or **t-SNE**
- ▶ see courses **Introduction to Artificial Intelligence** (Prof. d'Eramo) and **Data Mining** (Prof. Hotho)



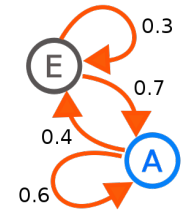
linear regression → L11

Outlook: Probabilistic graphical models

- ▶ in L14 and L15 we introduced **Markov models** as model for **stochastic processes**
- ▶ in L16 we considered **algorithmic applications** of Markov chains
- ▶ in L13 we considered **causal diagrams** where directed edges capture causal relationships

further topics

- ▶ **higher-order** Markov models
- ▶ **hidden Markov models** (HMM)
- ▶ **inference** for Markov chains
- ▶ **Bayesian networks** and **causal inference**
- ▶ temporal models: **Kalman filters** and **dynamic Bayesian networks**
- ▶ see courses **Data Mining** and **Time Series and Anomaly Detection** (Prof. Hotho)



graph representation of Markov chain with two states *A* and *E*

→ L13

image credit: Wikimedia Commons, User Joxemai4, CC-BY-SA

Notes:

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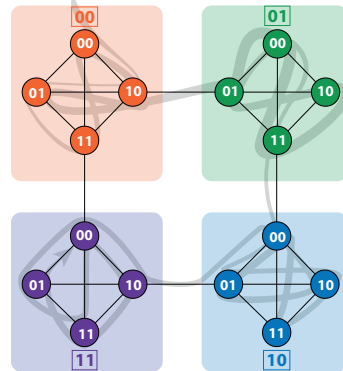
Outlook: Graph algorithms and graph learning

- ▶ in L18 we explored **cluster detection in graphs**
- ▶ in L19 studied graph Laplacians and **minimal cuts**
- ▶ in L20 we explored the Ford-Fulkerson method for **flow maximization** and proved the **max-flow min-cut theorem**

further topics

- ▶ **maximum matchings** in (bipartite) graphs
- ▶ **graph coloring** and labeling problems
- ▶ **graph drawing** algorithms
- ▶ **minimum spanning trees**
- ▶ **traveling salesman** problem (NP-complete)

- ▶ see course **Algorithmische Graphentheorie** (Prof. Wolff), **Statistical Network Analysis** and **Machine Learning for Complex Networks** (Prof. Scholtes)



`00|00|10|01|11|01|10|...01|00|01|10|11|00|...10|00|...`
flow compression with InfoMap → L17

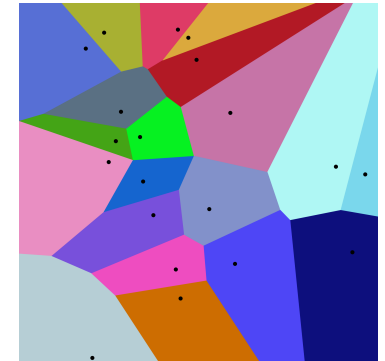
Outlook: Algorithmic geometry

- ▶ in our course, we did not consider specific **algorithms for geometric problems**
- ▶ algorithms for geometric problems are important in **computer graphics, robotics, and point cloud processing**

further topics

- ▶ **Voronoi diagrams**
- ▶ **Delaunay triangulation**
- ▶ **convex hull computation**
- ▶ **closest pair calculation**

- ▶ see course **Algorithmische Geometrie** (Dr. Klemz)



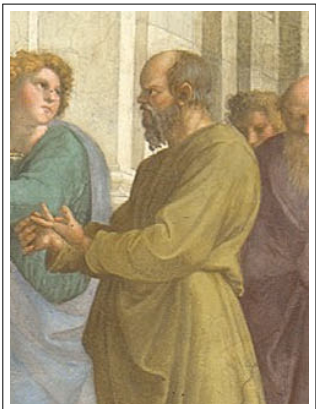
Voronoi diagram of a point cloud

image credit: Wikimedia Commons, Balu Ertl, CC BY-SA 4.0

Notes:

Notes:

Kindling the flame ...



Sokrates

image credit: Excerpt from Scuola di Athena, public domain



image credit: Wikimedia Commons, Andrew Dunn, CC BY-SA

“Education is the **kindling of a flame**, not the filling of a vessel.”

Sokrates

Advanced courses at JMU

Selected lectures

- ▶ Lecture: **Computer Vision**, Prof. Dr. Timofte
- ▶ Lecture: **Machine Learning for NLP**, Prof. Dr. Hotho
- ▶ Lecture: **Natural Language Processing**, Prof. Dr. Glavas, Prof. Dr. Hotho
- ▶ Lecture: **Theory of Machine Learning**, N.N.
- ▶ Lecture: **Information Retrieval**, Prof. Dr. Hotho
- ▶ Lecture: **Time Series and Anomaly Detection**, Prof. Dr. Hotho
- ▶ Lecture: **Machine Learning for Complex Networks**, Prof. Dr. Scholtes
- ▶ Lecture: **Statistical Network Analysis**, Prof. Dr. Scholtes
- ▶ Lecture: **Cognitive Systems**, Prof. Dr. N.N.
- ▶ Lecture: **Knowledge-based Systems**, Prof. Dr. Puppe
- ▶ Lecture: **Image Processing and Computational Photography**, Prof. Dr. Timofte
- ▶ Lecture: **3D Point Cloud Processing**, Prof. Dr. Nüchter
- ▶ Lecture: **Computational Complexity**, Prof. Dr. Wolff
- ▶ Lecture: **Algorithmic Graph Theory**, Prof. Dr. Wolff
- ▶ Lecture: **Algorithmic Geometry**, Dr. Klemz
- ▶ Lecture: **Optimization for Machine Learning**, N.N.

Seminars

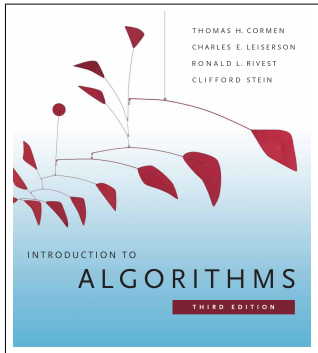
- ▶ Seminar: **Ausgewählte Kapitel des Machine Learning**, Prof. Hotho
- ▶ Seminar: **Aktuelle Trends in der Künstlichen Intelligenz**, Prof. Puppe
- ▶ Seminar: **Computer Vision**, Prof. Timofte
- ▶ Seminar: **Machine Learning for Complex Networks**, Prof. Dr. Scholtes
- ▶ Seminar: **Data, AI, and Society**, Prof. Dr. Scholtes
- ▶ Seminar: **Vision and Language**, Prof. Dr. Glavas and Prof. Dr. Timofte

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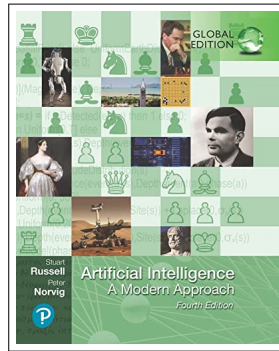
Notes:

Further reading ...

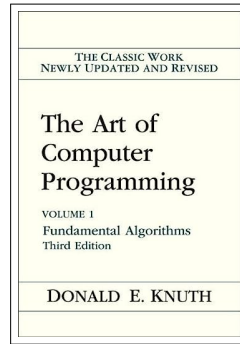
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TH Cormen, CE Leiserson, RL Rivest, C Stein: **Introduction to Algorithms**, 4th edition, 2022



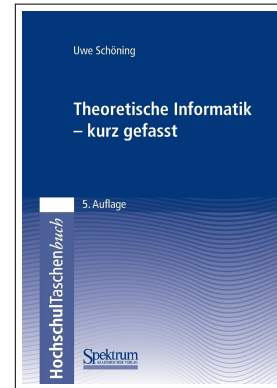
S Russel, P Norvig: **Artificial Intelligence: A Modern Approach**, 4th edition, 2020



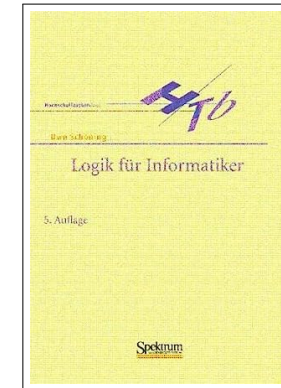
DE Knuth: **The Art of Computer Programming**, Vol. I – IV, 1968 – ?

Further reading ...

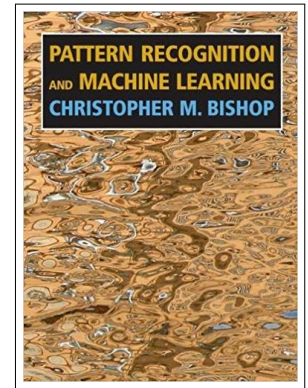
2/2



U Schöning: **Theoretische Informatik - kurz gefasst**, 5th edition, 2008



U Schöning: **Logik für Informatiker**, 5th edition, 2000



CM Bishop: **Pattern Recognition and Machine Learning**, 1st edition, 2006

Notes:

Notes:

Information on final exam

Q & A Session

today!

We answer questions that you identified during the exam preparation and offer to repeat selected parts of the lectures.

Final Exam

23.07.2025

- ▶ **Wednesday, July 23rd 2025**, from **14:30 - 16:30**, **Turing lecture hall**
- ▶ written, **closed-book exam** of 120 minutes
- ▶ no specific `python` questions (only “pseudocode” level)
- ▶ approx. 30 mins worth of **multiple-choice questions**
- ▶ approx. 90 minutes worth of open answer questions
- ▶ you can bring a **hand-written double-sided A4 “cheat” sheet**
- ▶ please **bring a copy of your exam registration** (registration number needed for exam cover sheet)



image credit: Randall Munroe, CC-BY-SA 2.5

Thank you!

- ▶ we covered basic foundations of **algorithms, AI, and data science**
- ▶ we **implemented and explored** them in `python`

some statistics

- ▶ **21 lectures** with **540 slides** generated by **33,255 lines of LaTeX code**
- ▶ **7200 lines of python code** producing **734 figures and illustrations**
- ▶ **32 group exercises**
- ▶ practice sessions with **42 jupyter notebooks**
- ▶ a **complete script is available** on WueCampus
- ▶ **thank you** for your attention and active participation!
- ▶ we hope to see many of you in **further courses, seminars, or your BSc thesis!**



Notes:

Notes: